

1 §1.3 Direction Fields

2 §1.3 Direction Fields (continued); Autonomous ODEs



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Lecture 3
§1.3 Direction Fields
Friday 11 September 2026

Announcements

- The TA is Illya Kierkosz (kierkosi@mcmaster.ca)
 - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: <https://davidearn.github.io/math2c/>
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
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Direction Fields

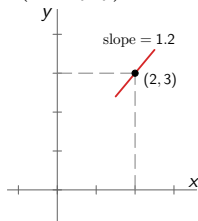
- Most nonlinear DEs cannot be solved exactly, *i.e.*, we cannot find an exact “quantitative” solution (we cannot find a formula for $y(x)$).
- But we can find approximate solution curves, using *qualitative methods*, without solving the problem exactly.
- A key qualitative tool is the *direction field* or *slope field*.

Direction Fields

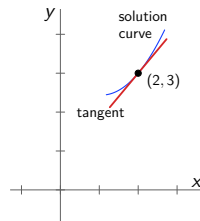
- **Recall:** If $f(x, y)$ and $\frac{\partial}{\partial y} f(x, y)$ are both continuous in a rectangle R , then there is a unique solution curve of $\frac{dy}{dx} = f(x, y)$ going through any point inside the rectangle R .
- At any point $(x, y(x))$ on a solution curve, the tangent line has slope $\frac{dy}{dx}$.
- We call a short segment of the tangent line through $(x, y(x))$ a **lineal element** at $(x, y(x))$.

e.g.,

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ &= 0.2xy\end{aligned}$$



(a) $f(2, 3) = 1.2$ is slope of lineal element at $(2, 3)$



(b) A solution curve passing through $(2, 3)$

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on Math 2C03
- Click on Take Class Poll
- Fill in poll **Slope of a lineal element**
- .

Direction Fields

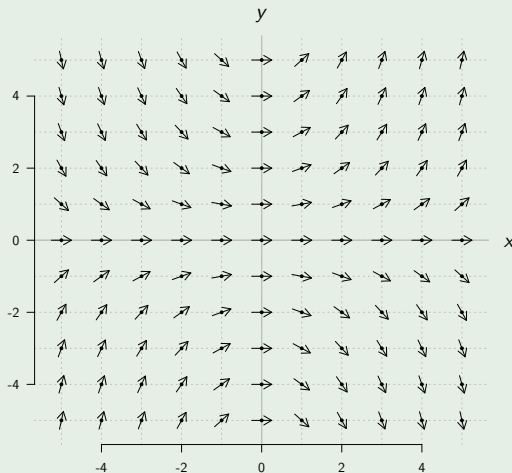
Definition (Direction Field)

A **direction field** of $\frac{dy}{dx} = f(x, y)$ is a collection of lineal elements plotted as arrows (pointing in the direction that the independent variable increases) on a grid of points in the plane.

- The direction field visually suggests the shape of a family of solution curves.

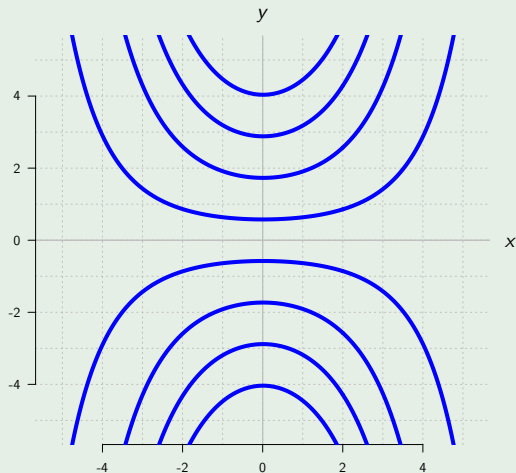
Direction Fields

Example (Direction field for $\frac{dy}{dx} = 0.2xy$)



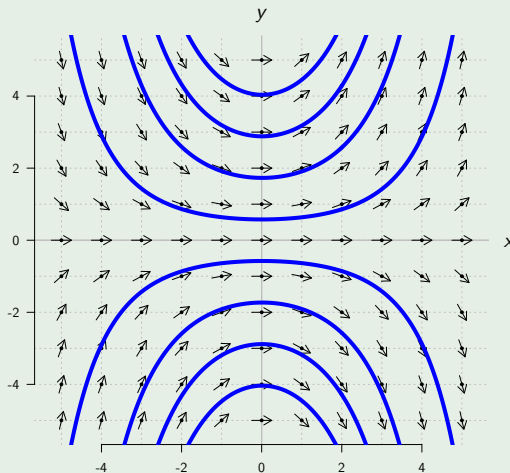
Direction Fields

Example (Solution curves for $\frac{dy}{dx} = 0.2xy$)



Direction Fields

Example (Direction field and solution curves for $\frac{dy}{dx} = 0.2xy$)



Direction Fields

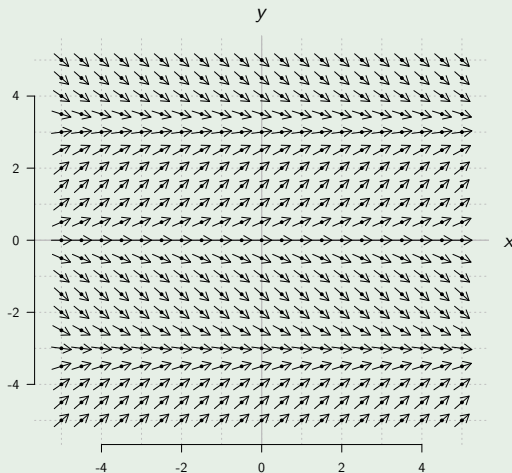
Now consider the first order ODE

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ &= \sin(y)\end{aligned}$$

- $f(x, y) = \sin(y)$ is continuous throughout the plane;
- $\frac{\partial}{\partial y} f(x, y) = \cos(y)$ is continuous throughout the plane;
- $\therefore$ there *is* a unique solution curve through every point in the plane.

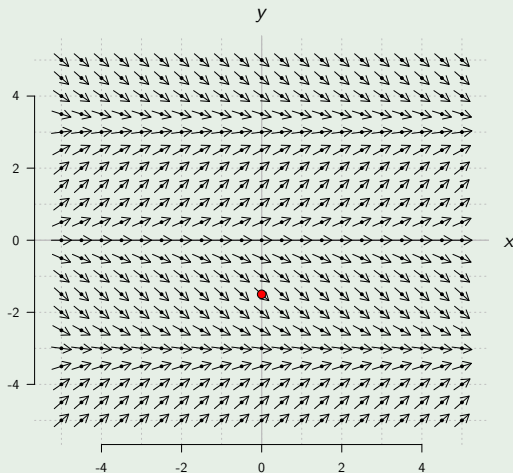
Direction Fields

Example (Direction field for $\frac{dy}{dx} = \sin(y)$)



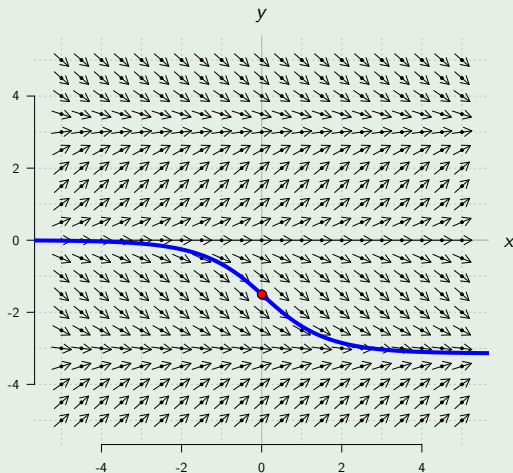
Direction Fields

Example (Direction field for IVP: $\frac{dy}{dx} = \sin(y)$, $y(0) = -\frac{3}{2}$)



Direction Fields

Example (Solution curve for IVP: $\frac{dy}{dx} = \sin(y)$, $y(0) = -\frac{3}{2}$)



Direction Fields

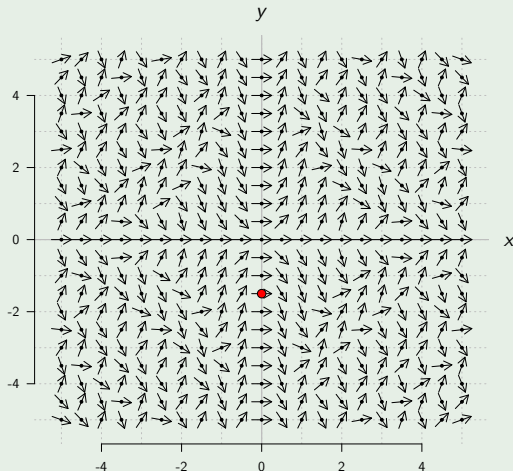
Now consider the first order ODE

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ &= 2\sqrt{1 + e^{-\sin(x^2+y^2)}} \sin(xy(1 + e^{-y^2}))\end{aligned}$$

- $f(x, y)$ is continuous throughout the plane;
- $\frac{\partial}{\partial y} f(x, y)$ is continuous throughout the plane;
- $\therefore$ there *is* a unique solution curve through every point in the plane.
- good luck finding exact solutions to this ODE...

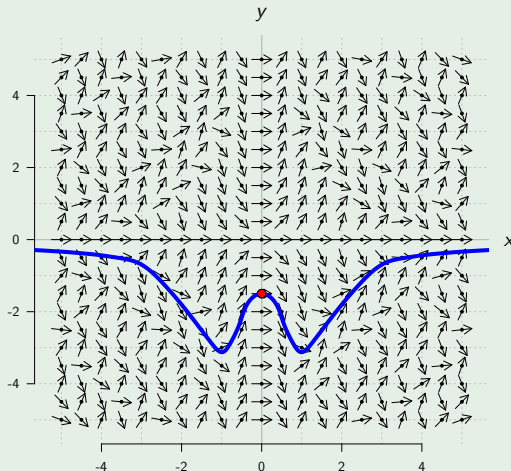
Direction Fields

Example (IVP: $\frac{dy}{dx} = 2\sqrt{1 + e^{-\sin(x^2+y^2)}} \sin(xy(1 + e^{-y^2}))$, $y(0) = -\frac{3}{2}$)



Direction Fields

Example (IVP: $\frac{dy}{dx} = 2\sqrt{1 + e^{-\sin(x^2+y^2)}} \sin(xy(1 + e^{-y^2}))$, $y(0) = -\frac{3}{2}$)





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and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Lecture 4

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Tuesday 15 September 2026

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Autonomous ODEs

An ODE is *autonomous* if the independent variable does not appear explicitly.

Example (Autonomous vs non-autonomous (first order) ODEs)

Autonomous:

$$\frac{dy}{dx} = \frac{1}{1+y^4}, \quad \frac{dy}{dt} = y e^{-y}$$

Non-autonomous:

$$\frac{dy}{dx} = \frac{1+x^4}{1+y^4}, \quad \frac{dy}{dt} = \cos(t) e^{-y}$$

Autonomous ODEs

The generic first order autonomous ODE is

$$F(y, y') = 0 \quad \text{general form}$$

$$\text{or} \quad \frac{dy}{dx} = f(y) \quad \text{normal form}$$

We will assume that $f'(y)$ exists and is continuous for the y -values in a rectangular region $R \subset \mathbb{R}^2$. Equivalently, $(x, y) \mapsto f(y)$ is continuously differentiable on R .

Note:

- $f(y)$ differentiable $\implies f(y)$ continuous
- $f'(y) = \frac{\partial f}{\partial y}$ because f is a function only of y
- $\therefore f'(y)$ continuous $\implies \frac{dy}{dx} = f(y)$ has a unique solution through every point inside $R \implies$ exactly one solution curve goes through any given point inside R .

Autonomous ODEs

A **critical point** or **equilibrium point** or **fixed point** of an autonomous first order ODE,

$$\frac{dy}{dx} = f(y),$$

is a value of y for which $\frac{dy}{dx} = 0$, i.e., a value of y for which $f(y) = 0$.

If c is a critical point, i.e., $f(c) = 0$, then

$$y(x) = c$$

is a **constant solution** or **equilibrium** of $\frac{dy}{dx} = f(y)$.

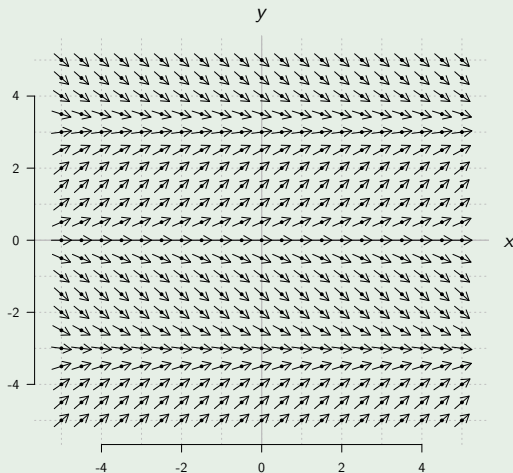
The solution curve of an equilibrium is a horizontal line in the xy -plane.

Poll

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- .

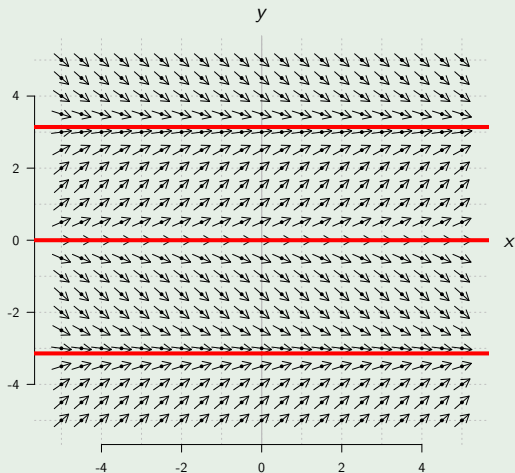
Autonomous ODEs

Example (Equilibria of $\frac{dy}{dx} = \sin(y)$)



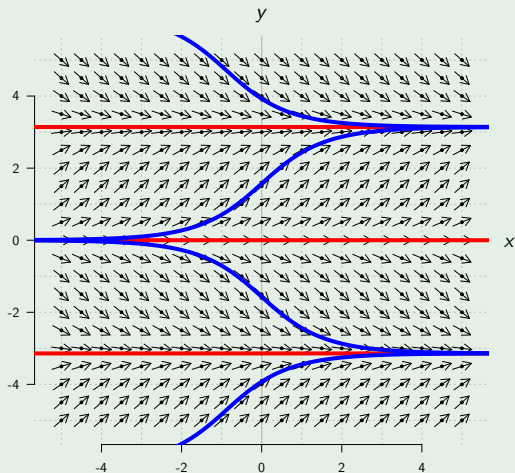
Autonomous ODEs

Example (Equilibria of $\frac{dy}{dx} = \sin(y)$)



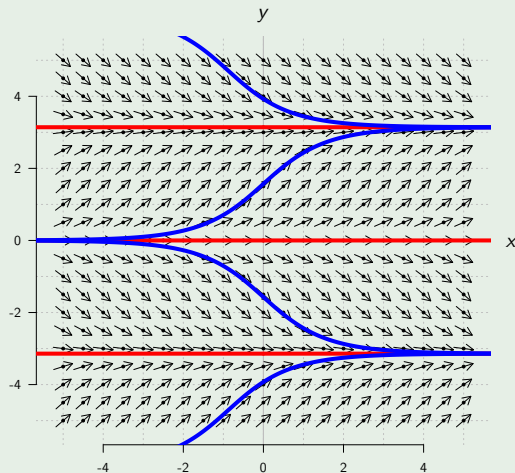
Autonomous ODEs

Example (Equilibria and other solution curves of $\frac{dy}{dx} = \sin(y)$)



Autonomous ODEs

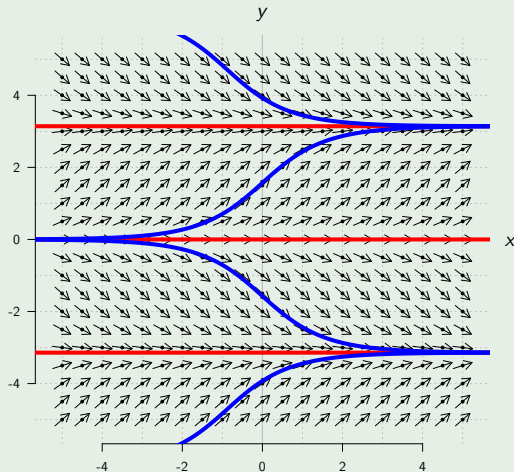
Example (Equilibria and other solution curves of $\frac{dy}{dx} = \sin(y)$)



Equilibria (red)
divide the xy -plane
into subregions

Autonomous ODEs

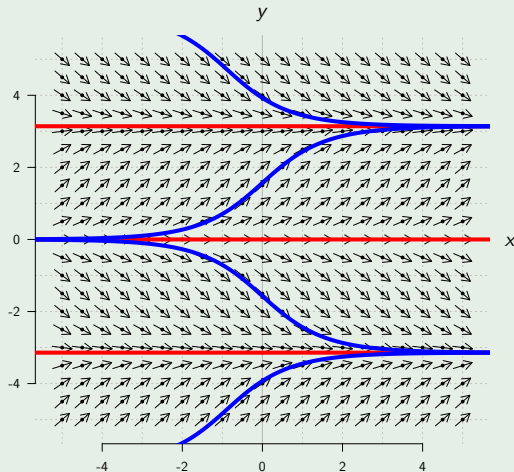
Example (Equilibria and other solution curves of $\frac{dy}{dx} = \sin(y)$)



Solution curves
(blue) never cross
between subregions

Autonomous ODEs

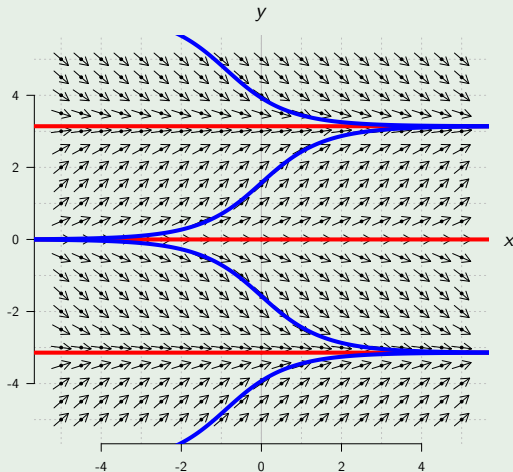
Example (Equilibria and other solution curves of $\frac{dy}{dx} = \sin(y)$)



Within a subregion,
 y either
strictly increases
or strictly decreases
as x increases

Autonomous ODEs

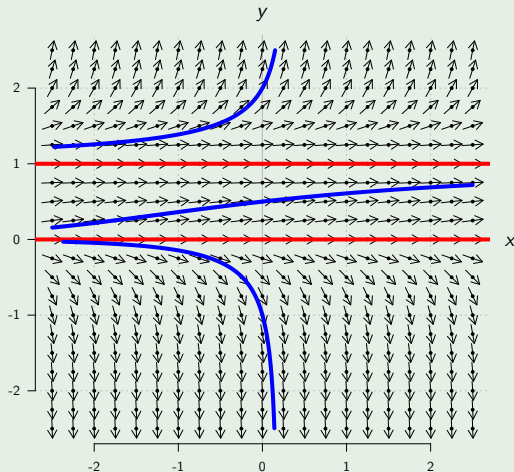
Example (Equilibria and other solution curves of $\frac{dy}{dx} = \sin(y)$)



For any solution curve (blue), as x increases, y approaches an equilibrium (red) (*in this example, not in general*)

Autonomous ODEs

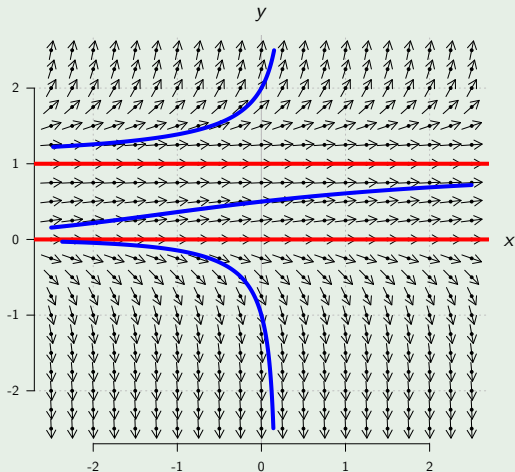
Example (Solution curves of $\frac{dy}{dx} = y(y-1)^2$)



$$\frac{dy}{dx} = y(y-1)^2$$

Autonomous ODEs

Example (Solution curves of $\frac{dy}{dx} = y(y-1)^2$)



■ Two equilibria:
 $y = 0$ and $y = 1$.

■ For $y > 1$, $\frac{dy}{dx} > 0$.

⇒ solution with $y(0) = 2$
increases and $y(x) > 2$
for all $x > 0$.

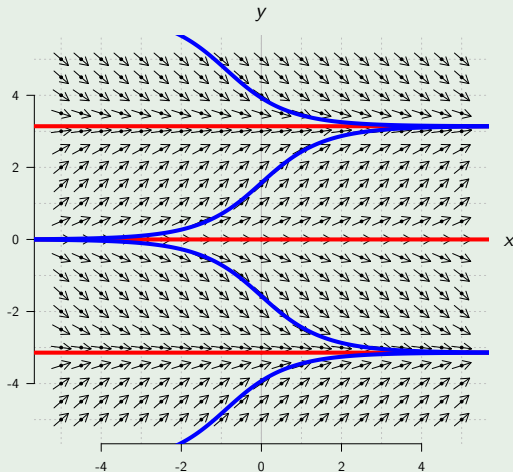
⇒ $\frac{dy}{dx} = y(y-1)^2 \geq 2$

⇒ $y(x)$ keeps increasing
at a rate at least 2

⇒ $y(x)$ cannot approach
an equilibrium as x
increases.

Autonomous ODEs

Example (Equilibria and other solution curves of $\frac{dy}{dx} = \sin(y)$)

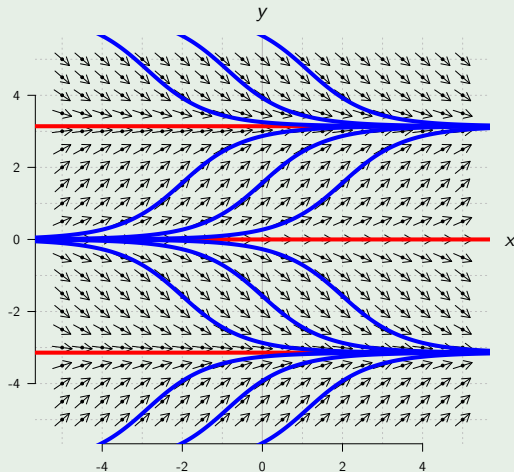


$$\frac{dy}{dx} = f(y) \implies$$

direction field does
not depend on x
 $\implies$ horizontal
translation of a
solution is another
solution

Autonomous ODEs

Example (Equilibria and other solution curves of $\frac{dy}{dx} = \sin(y)$)

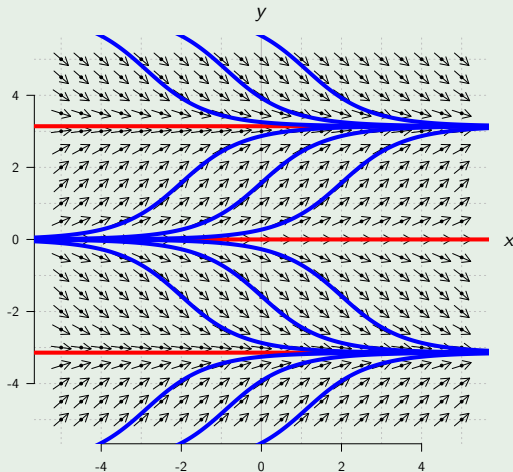


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Autonomous ODEs

Example (Equilibria and other solution curves of $\frac{dy}{dx} = \sin(y)$)

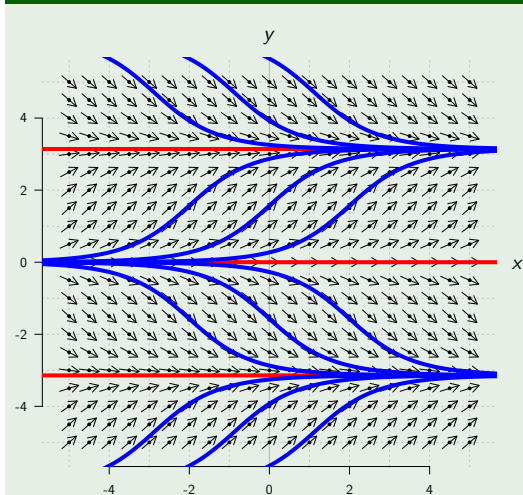


$$\frac{dy}{dx} = f(y) \implies$$

direction field does
not depend on x
 $\implies$ all information
about the direction
field can be obtained
from the y axis

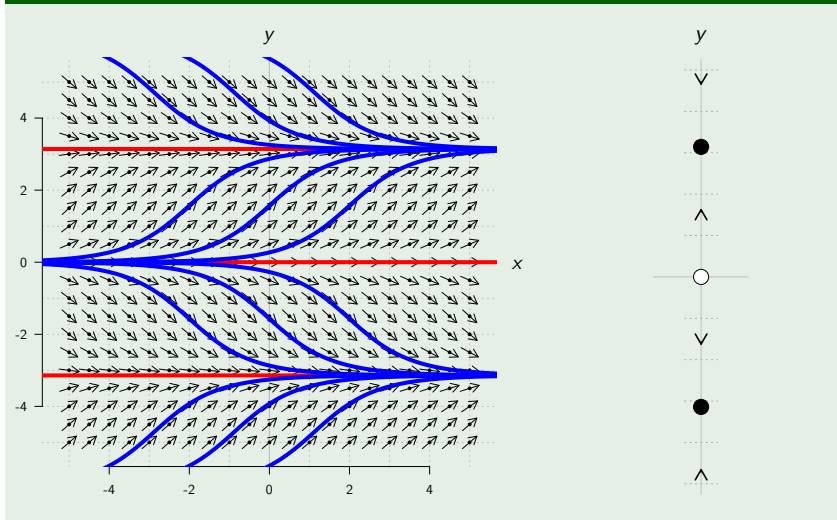
Autonomous ODEs

Example (Solution curves and phase line of $\frac{dy}{dx} = \sin(y)$)

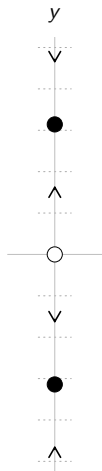


Autonomous ODEs

Example (Solution curves and phase line of $\frac{dy}{dx} = \sin(y)$)



Autonomous ODEs



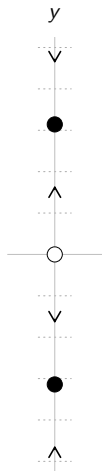
Phase Line

The *phase line* or *one-dimensional phase portrait* for an autonomous first order ODE,

$$\frac{dy}{dx} = f(y),$$

displays the direction in which the dependent variable y changes on solution curves.

Autonomous ODEs

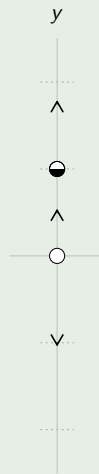
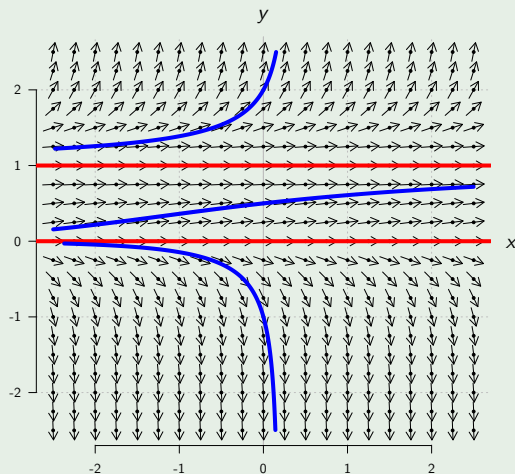


Phase Line

- Arrows indicate direction of change of the dependent variable (y) as the independent variable (x) increases.
- Black dots ● indicate *stable* equilibria
 - nearby points move toward ●
 - ● is an *attractor*
- White dots ○ indicate *unstable* equilibria
 - nearby points move away from ○
 - ○ is a *repeller*
- Half-filled dots ◐: *semi-stable* equilibria
 - motion depends on which side of ◐ or ◑

Autonomous ODEs

Example (Solution curves and phase line of $\frac{dy}{dx} = y(y-1)^2$)



Autonomous ODEs

Note: For systems of **two** autonomous first order ODEs, the **phase portrait** is **two**-dimensional.

- Two-dimensional systems of ODEs are a major focus in Math 3F03, which is mostly about qualitative analysis of nonlinear ODEs.
- Two-dimensional autonomous ODE systems can have attractors and repellers that are more complicated than equilibria.